MARKETING CAMPAIGN PERFORMANCE AND CUSTOMER BEHAVIOR ANALYSIS

1. Introduction

Marketing plays an important role in understanding customer needs and improving business performance. Analysing customer demographics, purchasing habits, and campaign responses helps businesses make better marketing decisions.

This project uses Python to analyse customer behaviour and marketing campaign performance. It focuses on spending patterns, product preferences, purchasing channels, and campaign acceptance.

Statistical hypothesis testing is also performed to compare the spending behaviour of customers who accepted the latest marketing campaign with those who did not.

2. Problem Statement

Businesses need to understand customer purchasing behaviour and marketing campaign effectiveness to improve customer engagement and marketing strategies.

This project analyses customer demographics, spending patterns, product preferences, and campaign responses to identify meaningful trends and generate actionable business insights.

3. Research Questions

1. Which product categories generate the highest customer spending?
2. Which purchase channel is most preferred by customers?
3. How do age and education levels relate to campaign response rates?
4. Which marketing campaign has the highest acceptance rate?
5. Do customers who accept the latest marketing campaign spend significantly more than customers who do not?

4. Project Objectives

- Clean and prepare the customer dataset.
- Check customer demographics and spending patterns.
- Identify top-performing product categories.
- Analyse customer purchasing channels.
- Evaluate marketing campaign acceptance rates.
- Compare campaign responses across customer groups.
- Perform statistical hypothesis testing.

- Generate business recommendations.

5. Dataset Description

Dataset Name: Marketing Campaign Results

Source: Maven Analytics Data Playground

Dataset Link: https://mavenanalytics.io/data-playground/marketing-campaign-results

Original Records: 2,240 rows

Original Columns: 28

The dataset contains customer information, including:

- Customer demographics
- Income and education
- Marital status
- Product category spending
- Purchase channels
- Previous campaign responses
- Latest campaign response

6. Tools and Technologies

TOOL	PURPOSE
PYTHON	Data analysis
PANDAS	Data manipulation
NUMPY	Numerical operations
MATPLOTLIB	Data visualization
SEABORN	Statistical visualization
SCIPY	Hypothesis testing
JUPYTER NOTEBOOK	Analysis environment
VS CODE	Project development

7. Data Cleaning and Preparation

The dataset was checked for missing values, duplicate records, incorrect data types, and suspicious values.

The following operations were performed:

1. Removed extra spaces from column names.
2. Converted customer registration dates into datetime format.
3. Checked missing values and duplicate records.
4. Identified three unrealistic customer ages and treated them as missing values.
5. Identified the suspicious income value of 666666 and replaced it with a missing value.
6. Filled missing income values using median imputation.
7. Grouped uncommon marital status categories into "Other".
8. Created new features such as Age, Total_Spending, and Total_Purchases.

Final cleaned dataset shape: 2,240 rows and 31 columns.

Three missing Age values were retained because they could not be reliably corrected.

8. Exploratory Data Analysis (EDA)

8.1 Customer Spending Analysis

The Total_Spending feature was created by combining spending across six product categories.

- Average customer spending: 605.80
- Median customer spending: 396
- Minimum spending: 5
- Maximum spending: 2,525

8.2 Product Category Analysis

Total spending by product category:

PRODUCT CATEGORY	TOTAL SPENDING
WINES	680,816
MEAT PRODUCTS	373,968
GOLD PRODUCTS	98,609
FISH PRODUCTS	84,057
SWEET PRODUCTS	60,621
FRUITS	58,917

Observation: Wines generated the highest spending, followed by Meat Products. Fruits generated the lowest spending.

8.3 Purchase Channel Analysis

PURCHASE CHANNEL	TOTAL PURCHASES
STORE	12,970
WEB	9,150
CATALOG	5,963
DEALS	5,208

Observation: Store purchases were the most frequently used purchasing channel, followed by web purchases.

8.4 Marketing Campaign Analysis

CAMPAIGN	ACCEPTED CUSTOMERS	ACCEPTANCE RATE
CAMPAIGN 1	144	6.43%
CAMPAIGN 2	30	1.34%
CAMPAIGN 3	163	7.28%
CAMPAIGN 4	167	7.46%
CAMPAIGN 5	163	7.28%
LATEST RESPONSE	334	14.91%

Observation: Campaign 4 had the highest acceptance rate among the five historical campaigns. The latest campaign response rate was 14.91%.

8.5 Age Group Analysis

The highest observed response rate was among customers aged 65+ (19.48%), followed by customers aged 26–35 (18.65%).

The 36–45 age group had the largest number of customers (758).

8.6 Education Analysis

EDUCATION	CUSTOMERS	CAMPAIGN RESPONSES	RESPONSE RATE
PHD	486	101	20.78%
MASTER	370	57	15.41%
GRADUATION	1,127	152	13.49%
2N CYCLE	203	22	10.84%
BASIC	54	2	3.70%

Observation: PhD customers had the highest observed campaign response rate, while Basic education customers had the lowest.

These are descriptive results and do not establish that age or education causes campaign acceptance.

9. Statistical Hypothesis Testing

Research Question

Do customers who accept the latest marketing campaign spend significantly more than customers who do not accept it?

Hypotheses

Null Hypothesis (H0): Customers who accept the latest marketing campaign do not have higher average spending than non-responders.

Alternative Hypothesis (H1): Customers who accept the latest marketing campaign have higher average spending than non-responders.

Test Used

Welch's Independent Samples t-test (One-tailed).

Results

METRIC	RESULT
RESPONDERS' AVERAGE SPENDING	987.39
NON-RESPONDERS' AVERAGE SPENDING	538.93
DIFFERENCE	448.46
T-STATISTIC	10.8503
P-VALUE	1.50×10^{-24}
SIGNIFICANCE LEVEL	0.05

Interpretation

The p-value is significantly smaller than 0.05. Therefore, the null hypothesis is rejected.

There is strong statistical evidence that customers who accepted the latest campaign had higher average spending than customers who did not accept it.

However, this result shows an association and does not prove that campaign acceptance caused higher spending.

10. Key Findings

1. Wines generated the highest total spending among product categories.

2. Store purchases were the most frequently used channel.
3. Campaign 4 had the highest historical campaign acceptance rate.
4. The latest campaign response rate was 14.91%.
5. Customers aged 65+ had the highest observed age-group response rate.
6. PhD customers had the highest observed education-group response rate.
7. Latest campaign responders spent 448.46 more on average than non-responders.

11. Business Recommendations

1. Study successful campaign strategies to improve lower-performing campaigns.
2. Focus on customer preferences for Wines and Meat Products.
3. Continue supporting store purchases while improving online engagement.
4. Use spending patterns to identify customers for loyalty and retention initiatives.
5. Test different marketing strategies for customer groups instead of relying only on demographic differences.
6. Evaluate campaigns using revenue, profit, conversion, and customer retention metrics.

12. Limitations

- The analysis is based on a single customer marketing dataset.
- Three unrealistic age values were retained as missing.
- Missing income values were filled using median imputation.
- The analysis identifies associations, not causal relationships.
- Campaign profitability and incremental campaign impact were not directly measured.
- Some customer groups have relatively small sample sizes.

13. Conclusion

This project demonstrates how Python-based data analysis and statistical hypothesis testing can help businesses understand customer behaviour and marketing campaign performance.

The analysis identified important trends in product spending, purchasing channels, customer demographics, and campaign acceptance.

The hypothesis test also showed a statistically significant difference in average spending between campaign responders and non-responders.

Overall, the project provides useful insights that can support data-driven marketing decisions.

Developed Using: Python, Pandas, NumPy, Matplotlib, Seaborn and SciPy

Dataset Source: Maven Analytics Data Playground